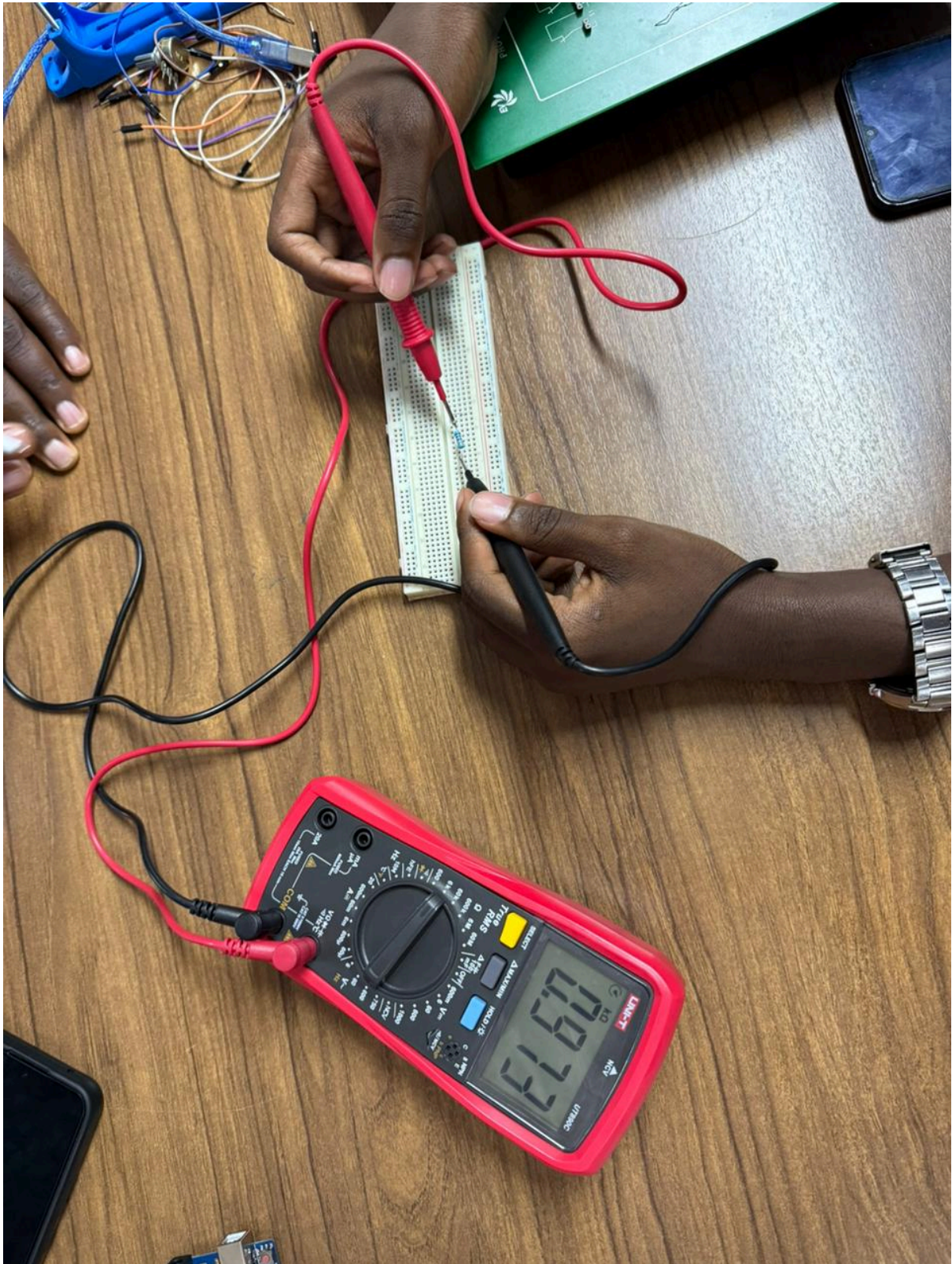


Group members: The Resistance



Experiment 1



Resistance measured using bands: $1000\Omega \pm 5\%$

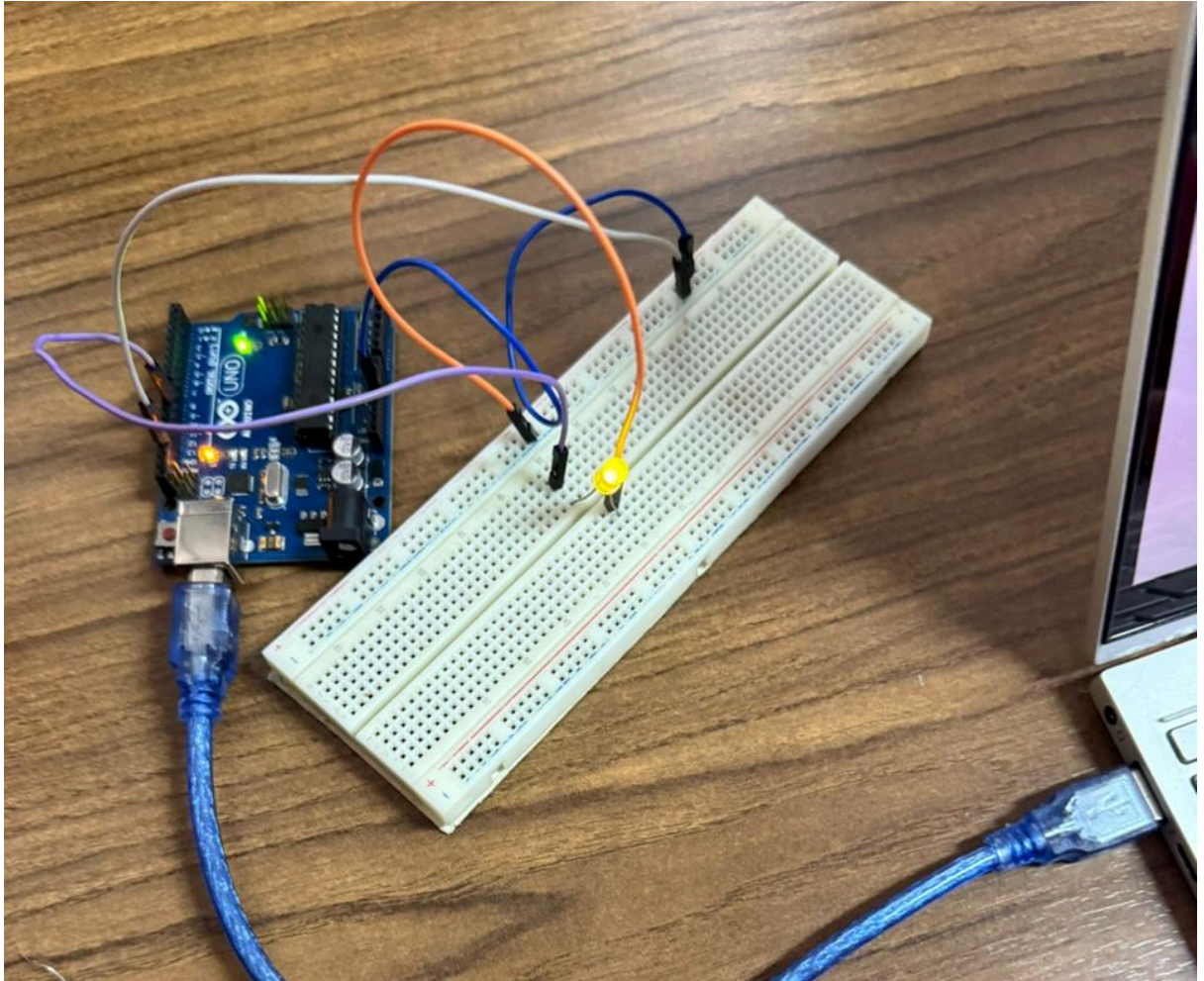
Resistance using multimeter: $0.973k\Omega$ (973Ω)

Experiment 1:

- Difference in values is because the gold band shows that these can be a variation of $\pm 5\%$.
- The difference observed is $1000\Omega - 973\Omega = 27\Omega$.
- The difference is 2.7% , which falls within the $\pm 5\%$.

Experiment 2

1. Physical implementation:

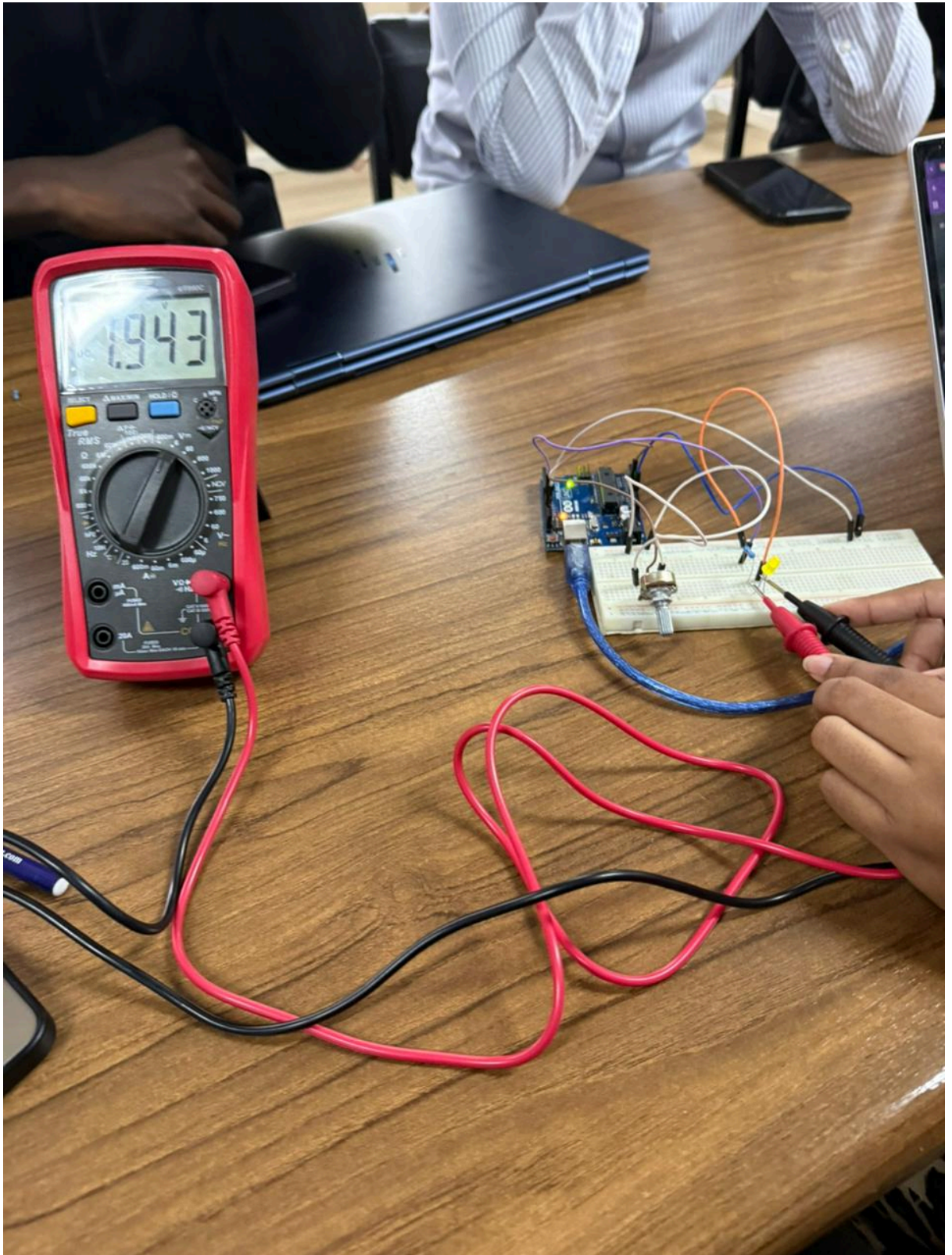


2. Blink program deployed:

```
void setup()
{
    pinMode(LED_BUILTIN, OUTPUT);
}

void loop()
{
    digitalWrite(LED_BUILTIN, HIGH);
    delay(1000); // Wait for 1000 millisecond(s)
    digitalWrite(LED_BUILTIN, LOW);
    delay(1000); // Wait for 1000 millisecond(s)
}
```


3. Voltage:



4. Pulse width modulation:

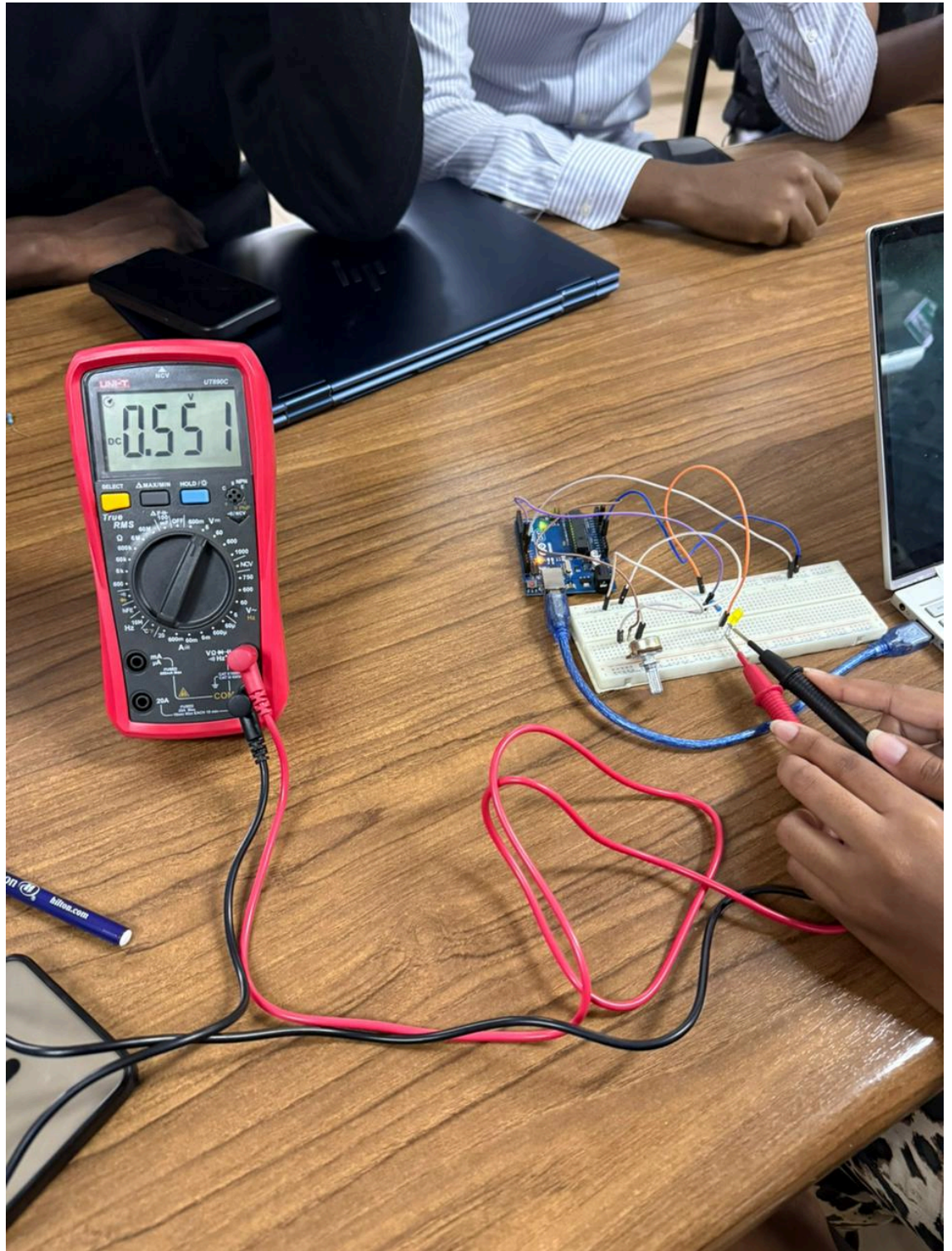
- Program used:

```
void setup()
{
  pinMode(9, OUTPUT);
}

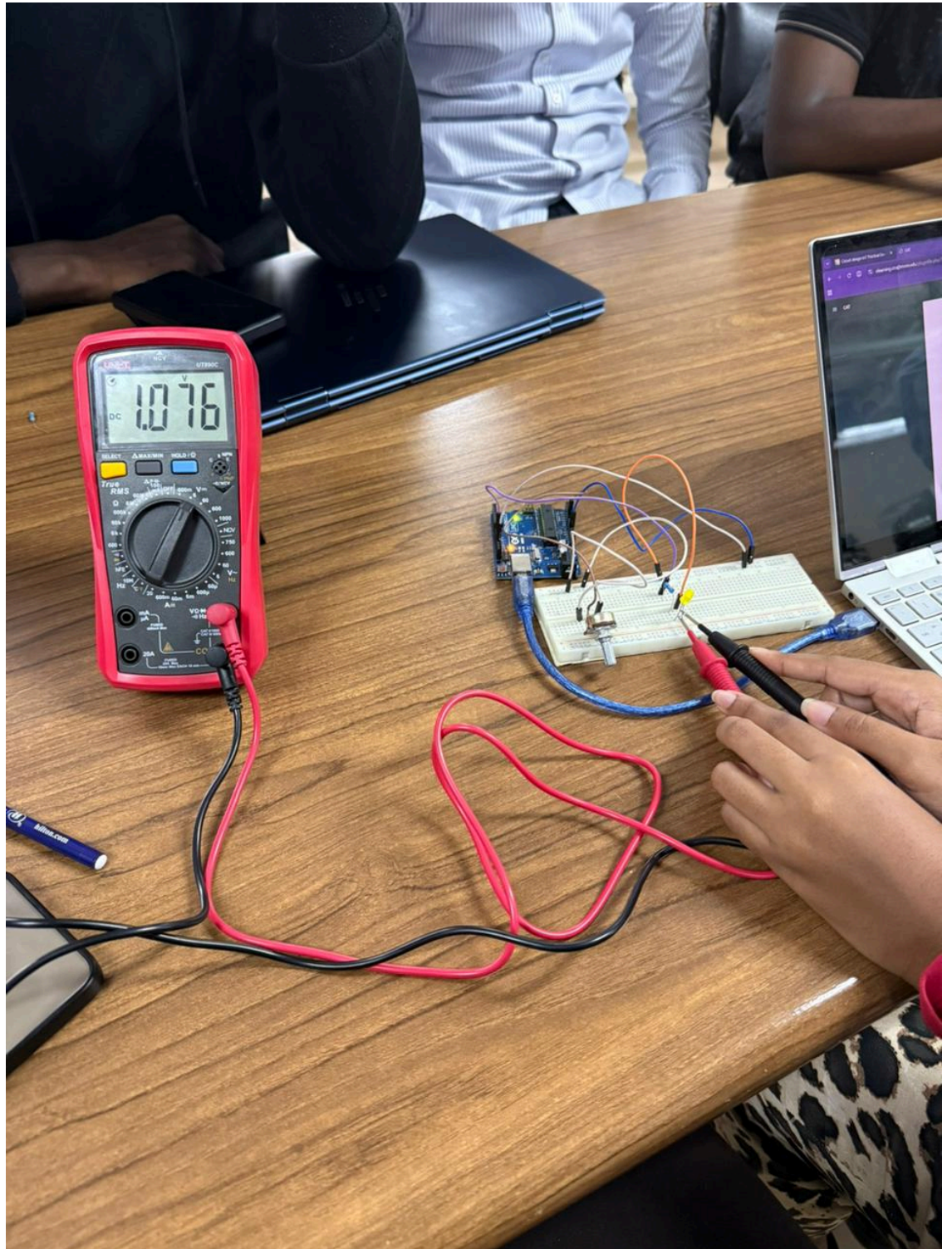
void loop()
{
  int potentiometer = analogRead (A0);
  int brightness = potentiometer / 4;

  analogWrite (9, brightness);
```


a. 25% brightness:



b. 50% brightness:



c. 100% brightness:

